Shirt Pipeline

Setup Guide: what to do on your PC

> **NEVER** Never paste your API keys or passwords into a chat, an email or a screenshot. They go in **one file on your own PC** (a file named **.env**) and nowhere else. You do **not** need to send me any key. The code is already pushed.

What you have now

- **Drop a photo, get a link.** It plans, makes and checks every layer, uploads to Google Drive and emails you the share link with a preview sheet. Hands-off is the default.
- **Automatic quality check.** Every layer is compared with your mockup (spelling, missing parts, stray elements, backdrop). Bad layers are redone automatically, up to 2 times.
- **The right style by itself.** Long sleeve becomes cut and sew (front, back, left sleeve, right sleeve). The log says why.
- **Claude can be the planner.** GPT only makes the images. You can switch either way with one line.
- **A web page** to drop mockups into, watch progress, copy the link, or redo one layer. It starts by itself when Windows starts.

The 10 steps

Step	What	Time	Needed?
1	Update the code on your PC	5 min	Yes
2	Install the new packages	2 min	Yes
3	Create your .env file	3 min	Yes
4	Get your Claude (Anthropic) API key	10 min	Yes, for Claude as planner
5	Set up email alerts (Gmail app password)	8 min	Recommended
6	Add your pattern diagram	1 min	Yes, for cut and sew
7	Google Drive on your PC, for phone uploads	10 min	Recommended
8	First run and first test shirt	10 min	Yes
9	Make it start by itself	1 min	Recommended
10	Use the web page from your phone	10 min	Optional

> **TIP** In a hurry? Do steps **1, 2, 3, 4, 6 and 8** to try everything today. Add steps 5, 7, 9 and 10 afterwards.

1 Update the code on your PC

about 5 min

Use the same method you used before. The newest code is on the branch **ccr-b7e053bd-hk7g02**.

Easiest: download a fresh ZIP of that branch from GitHub (**Code, Download ZIP**), unzip it, then copy everything from its **shirt-pipeline** folder into your working shirt-pipeline folder. Choose **Replace** for every file.

> **IMPORTANT** Do NOT overwrite or delete these, they are yours: **.env**, **credentials.json**, **token.json**, and the **Inbox**, **Ready** and **Work** folders. The ZIP does not contain them, so a normal copy keeps them.

2 Install the new packages

about 2 min

Open Command Prompt inside the shirt-pipeline folder (click the folder's address bar, type **cmd**, press Enter), then run:

```
pip install -r requirements.txt
```

This adds the Claude and web page packages. If it says pip is not recognized, use **python -m pip install -r requirements.txt**.

3 Create your .env file

about 3 min

The .env file holds your keys and settings. A template named **.env.example** is in the folder.

1. In File Explorer, find **.env.example** (turn on **View, Show, File name extensions** if you cannot see it).
2. If you already have a **.env** from before, keep it and just open it in Notepad (right-click, Open with, Notepad). Add the new lines from .env.example.
3. If you do not, copy .env.example, paste it, and rename the copy to exactly **.env** (no .txt on the end).
4. Open it in Notepad. Your OpenAI key goes on the line **OPENAI_API_KEY=** (no quotes, no spaces). Keep **IMAGE_MODEL=gpt-image-2.5-sunburst**.

> **TIP** A line that starts with **#** is a note and is ignored. Remove the # to switch a setting on.

4 Get your Claude (Anthropic) API key

about 10 min

This lets the pipeline use Claude as the planner and quality checker. It is **separate from a Claude.ai chat subscription** and is pay-as-you-go. Planning and checking one shirt typically costs well under a dollar.

1. Go to **console.anthropic.com** and sign in (or create an account).

2. Open **Settings, Billing** (or Plans and billing) and add a small amount of credit, for example $10.
3. Open **API keys**, click **Create key**, name it **Shirt Pipeline**, and copy the key (it starts with sk-ant-). You can only see it once.
4. In your .env file, paste it after **ANTHROPIC_API_KEY=** and confirm these two lines:

```
PLANNER_PROVIDER=claude
PLANNER_MODEL=claude-sonnet-5-5
```

> **TIP** Want to compare? Change the two lines to **PLANNER_PROVIDER=openai** and **PLANNER_MODEL=gpt-5.6-terra** and run the same shirt again. The quality check uses the same provider as the planner.

5 Set up email alerts

about 8 min

You get an email with the share link, the preview sheet and a list of anything that needs a look. This uses a Gmail **app password**, not your normal password.

1. Go to **myaccount.google.com** signed in as Cre8tiveminds242@gmail.com. Open **Security**.
2. Make sure **2-Step Verification** is turned on (turn it on first if it is not).
3. In the search box at the top of that page, type **App passwords** and open it. Name it **Shirt Pipeline** and click Create.
4. Google shows a 16-letter password. Copy it. Spaces are fine, the program ignores them.
5. Fill these three lines in .env:

```
NOTIFY_EMAIL_TO=Cre8tiveminds242@gmail.com
GMAIL_ADDRESS=Cre8tiveminds242@gmail.com
GMAIL_APP_PASSWORD=your16letterpassword
```

Then test it in Command Prompt:

```
python run.py test-email
```

A test email should arrive within a minute. If it says the login failed, the app password was copied wrong. Create a new one.

6 Add your pattern diagram

about 1 min

Your extraction rules use a long-sleeve pattern diagram for the cut and sew pieces. Save that image as exactly:

```
shirt-pipeline\rules\LONG_SLEEVE_PATTERN_REFERENCE.png
```

Without it the pieces are drawn from the size descriptions only, and the program tells you so. With it, the shapes follow your pattern.

7 Google Drive on your PC, for phone uploads

about 10 min

This lets you upload a shirt photo from your phone anywhere. It syncs to your PC, and the pipeline picks it up. **Your PC has to be on.**

1. Install **Google Drive for desktop** from google.com/drive/download and sign in as Cre8tiveminds242@gmail.com.
2. In Drive (on the PC), create a folder named **Shirt Inbox** inside My Drive.
3. Right-click that folder, choose **Offline access, Available offline** so files are really on the PC.
4. Find its path in File Explorer. It looks like **G:\My Drive\Shirt Inbox**. Put it in .env:

```
INBOX_DIR=G:\My Drive\Shirt Inbox
```

5. On your phone install the **Google Drive** app. To run a shirt, upload the photo into **Shirt Inbox**.

> **TIP** To add a correction from your phone, put it in the file name after two dashes: **roots shirt -- the animal on the left is a moose.jpg**. The part before the dashes becomes the shirt name.

8 First run and first test shirt

about 10 min

In Command Prompt, inside the shirt-pipeline folder:

```
python run.py serve
```

1. Leave that window open. Open **http://localhost:8765** in your browser.
2. Drag your **winter shirt** mockup onto the page. Leave the style on **Auto-detect**.
3. Watch the progress. Planning takes about a minute. The layers take a few minutes.
4. Check your email. You should get the share link and the preview sheet.
5. Look at the files in **shirt-pipeline\Ready\winter shirt**. The long sleeve should come out as **cut and sew** with four background pieces.

> **TIP** Layers marked amber or red on the preview sheet need a look. Click **Redo** on that layer on the web page. The Drive link stays the same.

Upscaling is still yours. The email lists the layers below the print target. Upscale them in Gigapixel, save them into the same **Ready** folder (the newest file with the same number wins), then click **Update Drive with my changes** on the page.

9 Make it start by itself

about 1 min

Stop the window from step 8 (press Ctrl+C), then run this once:

```
python run.py install-autostart
```

From now on the pipeline starts quietly every time you log in to Windows. It also puts a **Shirt Pipeline** shortcut on your desktop that opens the web page. If anything ever stops working, look at **Work\pipeline.log**. To undo it: **python run.py uninstall-autostart**.

10 Use the web page from your phone (optional)

about 10 min

The Drive upload in step 7 already covers phone use. This extra step lets you open the web page itself from your phone.

1. Install **Tailscale** (free for personal use) on the PC and on your phone, and sign in to the same account on both.
2. In .env, add these two lines. Pick your own password:

```
WEB_HOST=0.0.0.0
WEB_PASSWORD=choose-a-long-password
```

3. Restart the pipeline. On your phone, open **http://your-pc-name:8765** (the PC name is shown in the Tailscale app). Sign in with any user name and your password.

> **IMPORTANT** Always set WEB_PASSWORD when WEB_HOST is 0.0.0.0, and do not put this page on the open internet.

Everyday cheat sheet

I want to...	Do this
Run a shirt	Drop the mockup on the web page, or into the Inbox folder, or upload it to Shirt Inbox in Drive from my phone.
Add a correction	Type it in the page's Corrections box, or after **--** in the file name.
Force a style	Pick 3D all-over or Cut and sew on the page, or write **3D** or **cut and sew** in the corrections.
Hold one shirt for review	Tick **Hold this shirt for my review**, or add **--review** in the corrections.
Hold every shirt for review	Set **AUTO_APPROVE=0** in .env.
Redo one layer	Click **Redo** under that layer on the page.
Send my upscaled files to Drive	Save them in the Ready folder, then click **Update Drive with my changes**. Same link.
Spend less per shirt	Set **BEST_OF_HARD=1** (makes lettering and the main graphic once, not twice).

I want to...	Do this
See what a shirt cost	Open **Work\log.csv**. It lists images made and an estimated cost per shirt.

If something goes wrong

What you see	What to do
Email says **FAILED** and mentions a model that does not exist	Run **python run.py models**, then set **IMAGE_MODEL** in .env to a model on that list.
It says ANTHROPIC_API_KEY is empty	Step 4: paste your key after ANTHROPIC_API_KEY= in .env, then restart.
The web page will not open	Is the program running? Run **python run.py serve** and keep the window open, or check Work\pipeline.log.
Drive upload fails with an auth error	Delete **token.json** from the folder and run again. Sign in when the browser opens. (Google signs test apps out about every 7 days.)
A layer keeps coming out wrong	Click Redo, or add a correction naming the problem and run the shirt again.
Wrong style chosen	Add **3D** or **cut and sew** to the corrections. Never write 3D for a shirt that should be cut and sew.
No email arrives	Run **python run.py test-email**. Check spam. Re-create the app password if the login fails.

IMPORTANT The quality check is another AI. It catches most spelling mistakes and stray elements, but not everything. Glance at the preview sheet before you send a link to a customer.

What to send me after your first run

- A screenshot of the web page when the shirt finishes.
- The preview sheet from the email.
- The text from the **Show log** panel on the page, or from **Work\log.csv**.
- Anything that looked wrong, and what you expected instead.

NEVER Do not send keys, passwords, the .env file or token.json.